Lesson 33: Finding and Evaluating Sources

English Stage 3 (C1) · Unit 4: Research and Evidence

Adapted from Reading, Writing, Research, and Reasoning: An Advanced ESL Text by Gabriel Winer and Elizabeth Wadell, ASCCC Open Educational Resources Initiative (CC BY-NC 4.0): 3.6 Strategies for Finding Information; 3.7 Evaluating Your Sources.

In this lesson

You compare types of sources, turn a research question into keywords, and search the internet and free academic databases instead of a college library. Then you learn six criteria for evaluating sources and find and evaluate the first sources for your project, a core skill for university research.

Learning goals

After this lesson you should be able to:

- Compare types of sources by how they are created, reviewed, written and read.
- Choose background sources and research sources for different purposes.
- Turn a research question into keywords, synonyms, narrower terms, related terms and subject terms.
- Search Google Scholar and free databases, and make a search narrower or broader.
- Evaluate a source for authority, accuracy, relevance, currency, objectivity and style.

Key language

Word or phrase	Meaning	Example
search engine	a program that searches the internet	Google, Baidu, Bing
database	a collection of articles, usually paid for by a library, with each article tagged by keywords	Academic Search Complete
keyword	an important word (usually a noun) for a main concept of your topic	"students of color", "professors of color"
synonym	a word with the same or a similar meaning	minority faculty, faculty of color
subject term	a standard word or phrase that a database uses for the content of an article	"college teachers" (for "professors")
record	the page about one article in a database: author, publication, abstract (summary), subject terms	an ERIC record
peer review	experts in the field check an article before it is published	Scholarly journal articles have a formal peer review process.
narrow / broaden a search	make it more specific (fewer results) / more general (more results)	"community college students" / "students"

Word or phrase	Meaning	Example
current, currency	recent; how recent the information is	Is currency pertinent to your research?
bias, biased	a one-sided view; showing only one point of view	Is the author pretending to be objective but trying to sell something?
funder	a person or organization that pays for research	The website lists who funded the research.

Rules and explanations

Types of sources (Table 3.6.1)

The table in the lesson material compares sources by where you find them, the time to create them, the review process, the writers and the readers. The key difference: only **scholarly journal articles** have a formal peer review process; they take one to a few years to create and are written by specialists (usually people with PhDs, graduate students or professors) for scholars and graduate students. Most other sources are written for the general public, with editorial review or no review at all.

Which source for which purpose?

- **Background** at the start of a project: Wikipedia, newspaper articles, encyclopedias (in psychology, also a free textbook such as OpenStax *Psychology 2e*).
- **Proof** of an idea in your paper: scholarly journal articles, which show the results of original research. They are very specific, so they are not good for background.

Internet and database searching

- A web search gives millions of results (Lily got over 400 million), and the first page is not usually the most credible. Use it to understand your topic and find keywords; Google's "Tools" can limit results by date or country.
- A library database is not free. It collects articles (one journal article can cost $25) and tags each with keywords. Search more than one. Drawbacks: new information appears late, and some topics are not covered.

Without a college library, use these free tools (links in "Watch online"): | Free tool | What it is | Useful features | |---|---|---| | Google Scholar | a search engine for scholarly articles and books | date filter on the left ("Since ..."); free [PDF] links on the right; "Cited by" shows how often others cite a work | | PubMed Central | free full-text articles in medicine, health and the life sciences, including much psychology | an article's PubMed record lists its MeSH terms (subject terms) | | ERIC | a database of education research | "Peer reviewed only" and "Full text available on ERIC" boxes; each record lists Descriptors (subject terms) | | DOAJ | the Directory of Open Access Journals | every article is free to read | | OpenStax *Psychology 2e* | a free introductory psychology textbook | background information and key terms |

Keywords and subject terms

- Google guesses what you mean from a whole question; databases do not. Type only **keywords**: nouns for the main concepts, without verbs, adjectives, adverbs, prepositions or words like *affect/affected*.
- Book example: "How are students of color affected by professors of color?" Keywords: "students of color", "professors of color".
- For each concept, list **synonyms** (minority students; minority faculty, minority professors), **narrower terms** (African American students, community college students) and **related terms** (retention, educational achievement). Type each concept in its own search field.
- Too few results: use more general words. Too many: use more specific words or filters (date, type of article, language, geography).
- Find synonyms in a dictionary or thesaurus (such as Longman), in news or general websites, and in **subject terms** on an article's record page. Databases may use different terms for the same idea ("stadium", "sports arena").

Evaluating sources (Table 3.7.1)

Every source comes from a person or group with a motive, and every author has an opinion. | Criterion | Questions to ask | |---|---| | Authority/Credibility | Who is the author, and is there evidence of expertise? Are there reputable citations? | | Accuracy | Can the facts be verified in other sources? Are there spelling or grammar errors? | | Relevance | Does it answer your research question? Is it popular or scholarly? | | Currency/Date | When was it published or updated? Does currency matter for your topic? | | Objectivity/Bias/Reliability | What is its purpose (educational, commercial, promotional ...)? Is it pretending to be objective while selling or persuading? | | Style/Functionality | Is it well written and organized? Does a website work, without broken links? |

The book's **Source Evaluation Checklist** (your tutor has it) asks Yes/No questions on credibility, relevance, date, audience and purpose. A "no" means look more closely, not always reject. Careful: for "Is the information in this source biased or limited to one point of view?", the warning answer is "yes". (Its last line says "quality of your service"; it means "source".)

> **Psychology note:** In psychology, the strongest evidence usually comes from peer-reviewed journal articles that report empirical studies (studies based on observation or experiment). The main psychology database, APA PsycInfo, needs a library subscription, so use Google Scholar and PubMed Central.

Practice

Exercise 1. Types of sources. Use Table 3.6.1 in the lesson material.

1. Which type of source has a formal peer review process?
2. Which type can be created in seconds?
3. Who writes reports and government documents, and where can you find them?
4. Which three types can have editorial review but no peer review?

5. Who reads scholarly journal articles?
6. The book asks: does the time it takes to create a source affect its quality? Give your opinion with a reason.

Exercise 2. Background or proof? Which kind of source would you use first?

1. You know little about sleep and want a general overview.
2. You want to prove that sleeping after studying helps students remember.
3. You need news about a university that changed its exam times last week.
4. You want official numbers on how many students attend college in the United States.

Exercise 3. Keywords. Write the keywords for each research question.

1. How does sleep affect memory in university students?
2. Why do international students experience high levels of stress?
3. How can parents help their children maintain a heritage language?
4. What are the most significant effects of social media use on teenagers' self-esteem?
5. For question 1, list synonyms, narrower terms or related terms for each keyword.

Exercise 4. Which criterion? Match each situation with a criterion from Table 3.7.1.

1. The article was published in 1993, and your paper is about teachers today.
2. A website lists the foundations that paid for its research.
3. The author is a professor at the University of Illinois at Chicago.
4. Several links on the website are broken.
5. A blog gives a statistic that is different from the government's numbers.
6. The article is about elementary school teachers, but your paper is about college faculty.
7. A website about sleep sells sleeping pills and says that its pills improve memory.

Exercise 5. Try this! Evaluating some sources (3.7). Read the citation, author information and abstract of Source 1 (King) and Source 2 (Carver-Thomas) in the lesson material. Evaluate each with Table 3.7.1. Would it be a good source for a research paper on whether colleges need more faculty of color?

Exercise 6. Apply this! Your keyword worksheet (3.6). Make a worksheet like Lily's (Table 3.6.2): research question; keywords; synonyms from a dictionary; related terms from articles; subject terms from databases (search ERIC or PubMed, open a record, and copy its Descriptors or MeSH terms).

Speak with your tutor

- "Act as a librarian. Here is my research question: [paste]. Help me choose keywords, synonyms and narrower terms, then tell me which free database to try first."
- "I searched Google Scholar for [keywords] and got [number] results. Help me narrow or broaden my search."

- "Here is the citation and abstract of a source I found: [paste]. Ask me questions for each of the six criteria in Table 3.7.1, then tell me if my evaluation is fair."
- "Describe three sources on sleep and memory, one strong and two weak. I will evaluate them; then tell me what I missed."

Read aloud

Stressed syllables are in CAPITALS.

1. The results on the first page of a search / are not usually the most credible sources.
2. Select nouns that are important / and leave out words that are adverbs, verbs, adjectives, and prepositions.
3. If you're getting too many results, / use more specific words.
4. The organization may not be unbiased, / but it makes it clear who is funding the research.
5. Check the author's cred·i·BIL·i·ty / before you use the source.

Writing task

Find and evaluate the first sources for your project (your own topic, sleep and memory in university students, stress and coping among international students, or the book's topic, faculty of color).

1. Complete your keyword worksheet (Exercise 6).
2. With the free tools, find four to six sources: one background source, at least two scholarly articles with free full text, and one popular source. Note the author, title, publication, date and link or DOI of each.
3. Evaluate two of them (100 to 150 words each), like the book's answers for Sources 1 and 2: the author's credibility, accuracy, relevance, date and possible bias, then a decision: is it a good source for my paper, and why?

Self-editing checklist:

- Did I name the author and describe the author's background?
- Did I say what kind of source it is (scholarly or popular) and where it was published?
- Did I check the date and say whether currency matters for my topic?
- Did I consider purpose, funders and possible bias?
- Did I explain how the source relates to my research question?
- Did I record all the details I will need for my citation?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "How Google Search Works (in 5 minutes)", the video in 3.6: https://www.youtube.com/watch?v=0eKVizvYSUQ
- Google Scholar: https://scholar.google.com

- PubMed Central: https://pmc.ncbi.nlm.nih.gov
- ERIC: https://eric.ed.gov
- Directory of Open Access Journals: https://doaj.org
- OpenStax *Psychology 2e*: https://openstax.org/details/books/psychology-2e
- Source 2 in Exercise 5, free online: https://learningpolicyinstitute.org/product/diversifying-teaching-profession-brief

Answer key

Exercise 1

1. Scholarly journal articles.
2. Social media (seconds to days).
3. Specialists (usually people with PhDs, graduate students or professors); on .gov websites and organization websites.
4. News and magazines (sometimes editorial review) and books (editorial review).
5. Scholars and graduate students.
6. (example answer) "Usually yes: research and peer review take time, so a scholarly article is usually more reliable than a quick social media post. But time alone does not guarantee quality."

Exercise 2

1. A background source: Wikipedia, an encyclopedia, newspaper articles, or a textbook such as OpenStax *Psychology 2e*.
2. Scholarly journal articles (original research).
3. A news article: it is current, while scholarly articles take a year or two to be published.
4. Reports and government documents (.gov websites), for example from the National Center for Education Statistics.

Exercise 3

1. sleep; memory; university students.
2. international students; stress.
3. parents; children; heritage language.
4. social media; teenagers; self-esteem (leave out "most significant effects", as you leave out "affect").
5. (example answers) sleep: sleep deprivation, sleep loss, sleep duration, sleep quality, naps; memory: learning, recall, memory consolidation, academic performance; university students: college students, undergraduates, first-year students.

Exercise 4

1. Currency/Date.
2. Objectivity/Bias/Reliability (it shows who might influence the research).
3. Authority/Credibility.
4. Style/Functionality.
5. Accuracy.
6. Relevance.
7. Objectivity/Bias/Reliability: its purpose is commercial, so it may be selling while it seems objective.

Exercise 5 (book)

- Source 1: Dr. King seems highly credible, and her article was published in an academic journal on education. It discusses the benefits of having African American teachers and the problems with keeping them, which is relevant. However, it was published in 1993: it would be useful for comparing teacher diversity now and in the past, but not strong for the experiences of African American teachers today. It is also not about college teachers.
- Source 2: The research was done by an organization, not a university professor. The organization states its goals clearly and lists its funders, so even if it is not unbiased, it is clear who pays for the research. The research is relevant and fairly current. It may be a good source for a paper on whether it is important to have teachers of color in colleges.
- Note: the book's key does not mention that Source 2, like Source 1, is about school teachers, not college faculty (its abstract mentions teacher licensure, residencies and high-need schools). It is useful for the general benefits of teachers of color, but only indirectly relevant to colleges.

Exercise 6 (example answer, the guides' psychology project) | Strategy | Example | |---|---| | My research question | How does the amount of sleep affect memory and learning in university students, and what can students do to sleep better? | | Keywords | sleep, memory, university students | | Synonyms from a dictionary | rest (sleep); recall (memory); college students | | Related terms from articles I read | sleep deprivation or sleep loss or sleep quality; memory consolidation or learning or academic performance; undergraduates | | Subject terms in databases | PubMed MeSH: Sleep Deprivation, Memory Consolidation, Students; ERIC Descriptors: Sleep, Memory, College Students |

My notes